

Public Finance

BIEM – Spring 2026

Externalities and Public Goods (Ch.5 & 7 & 9.1)

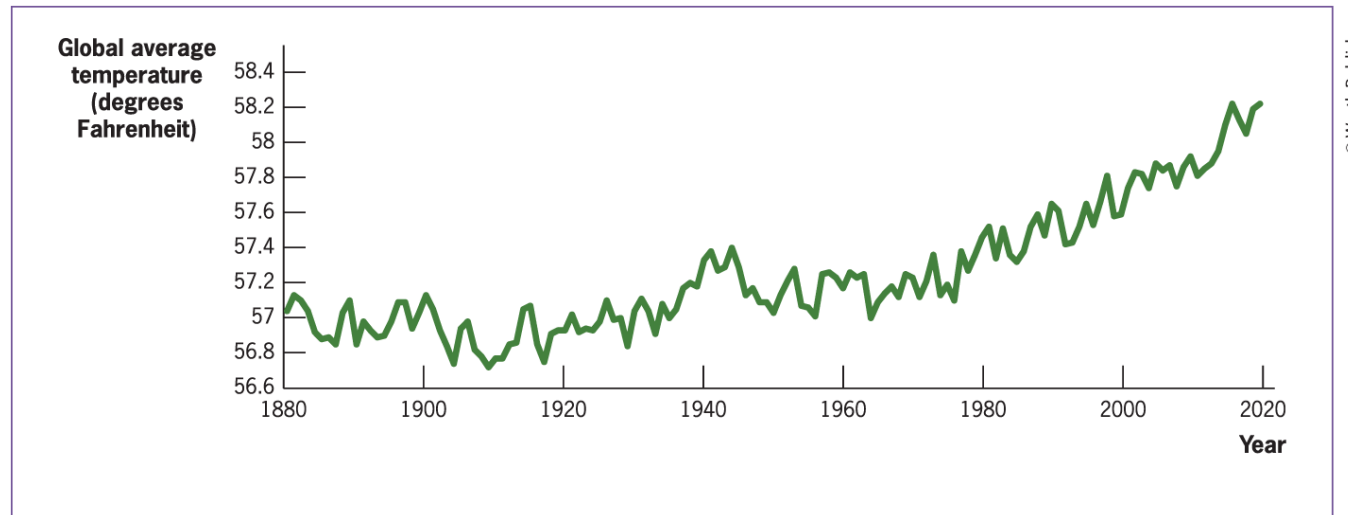


Externalities: in this course

- 5.1 Externality Theory (quick recap)
- 5.2 Private-Sector Solutions to Negative Externalities



Introduction to Externalities: Global Warming



- In 2015, representatives from 195 nations met in Paris, France, to negotiate an international pact to limit temperature rise around the world.
- Carbon dioxide emissions contribute to global warming, which could cause enormous damage.
- The cost of reducing the use of fossil fuels, particularly in the major industrialized nations, is immense. Some predict that we will have to reduce our use of fossil fuels to nineteenth-century (preindustrial) levels.



Externalities: Problems and Solutions

- Global warming is a classic example of an externality, which is a kind of market failure.
- **Externality:** Externalities arise whenever the actions of one party make another party worse or better off, yet the first party neither bears the costs nor receives the benefits of doing so.
- **Market failure:** A problem that causes the market economy to deliver an outcome that does not maximize efficiency.



There are four possible types of externality:

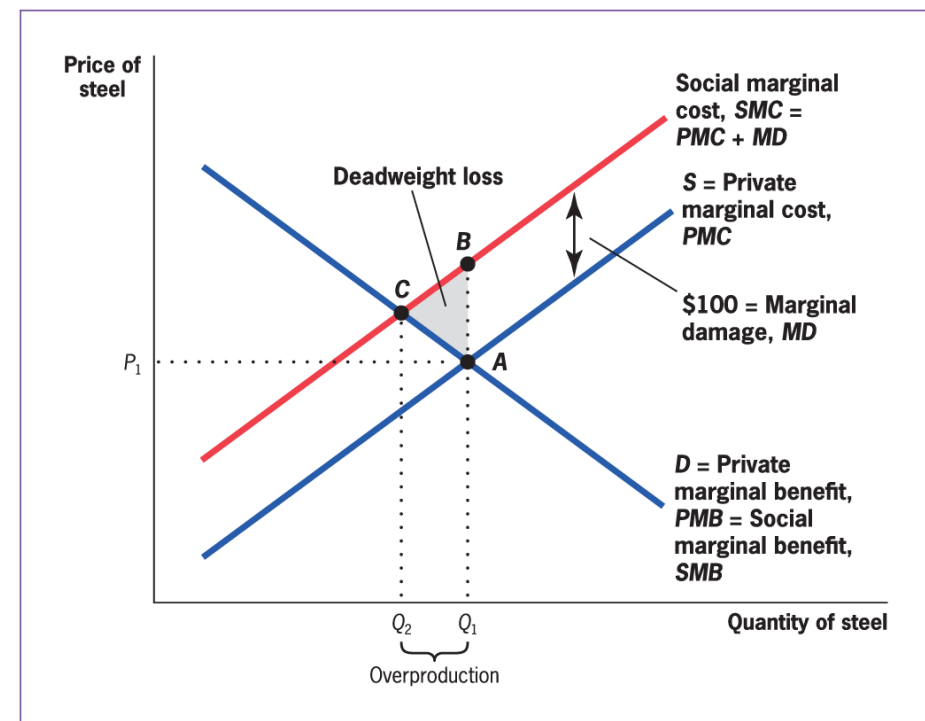
	Production	Consumption
Positive	SMC curve lies below PMC curve Ex: investment in R&D	SMB curve lies above PMB curve Ex: vaccines
Negative	SMC curve lies above PMC curve Ex: steel production curve	SMB curve lies below PMB curve Ex: smoking



An example: Negative production externality

Negative production externality:

- When a firm's production reduces the well-being of others who are not compensated by the firm (Ex: pollution from steel production)
- They drive a wedge (marginal damage, MD) between private and social marginal cost.
 - Private marginal cost (PMC): The direct cost to producers of producing an additional unit of a good.
 - Social marginal cost (SMC): The private marginal cost to producers plus any costs associated with the production of the good that are imposed on others.



There is overproduction of $Q_1 - Q_2$, with an associated deadweight loss of area BCA .

Private-Sector Solutions to Negative Externalities: The Solution

- Externalities undermine efficiency because one party does not pay the costs or get all the (net) benefits of its actions.
- The solution to this, therefore, is to **internalize the externality**.
- Internalizing the externality: When either private negotiations or government action led the party to fully reflect the external costs or benefits of that party's actions.
 - Steel producer could pay the fishers for each unit of steel produced so that they are compensated for the damage to their fishing grounds.



The Solution: The Coase Theorem

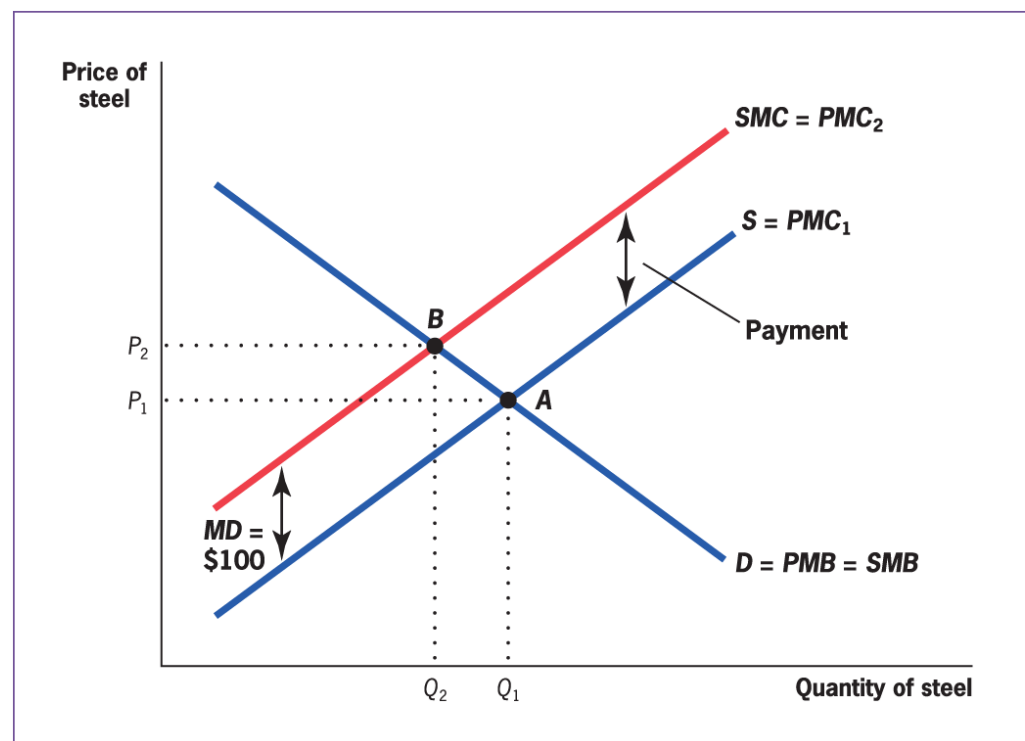
The Coase theorem says that private parties will be able to solve the problem of externalities. This is accomplished by internalizing the externality.

- Coase theorem (Part I): When there are well-defined property rights and costless bargaining, then negotiations between the party creating the externality and the party affected by the externality can bring about the socially optimal market quantity.
- Coase theorem (Part II): The efficient solution to an externality does not depend on which party is assigned the property rights as long as someone is assigned those rights.



The Solution: Coasian Payments

- The charge internalizes the externality and removes the inefficiency of the negative externality.



The Problems with Coasian Solutions 1

- There are difficulties with Coasian solutions, making them less likely to arise as more people become involved.
- The assignment problem: The first problem is assigning blame. Does the fisher pay the steel plant for not polluting? Or does the steel plant pay for polluting?
- The holdout problem: Shared ownership of property rights gives each owner power over all the others. Each person has veto power and so may demand enormous payments.



The Problems with Coasian Solutions 2

- The free rider problem: When an investment has a personal cost but a common benefit, individuals will underinvest. Individuals may not want to pay enough to reduce pollution.
- Transaction costs and negotiating problems: It is hard to negotiate when there are large numbers of individuals on one or both sides of the negotiation.
 - This problem is amplified for an externality such as global warming, where the potentially divergent interests of billions of parties on one side must be somehow aggregated for a negotiation.



Bottom Line

- Ronald Coase's insight that externalities can sometimes be internalized was a brilliant one.
- It provides the competitive market model with a defense against the onslaught of market failures.
- It is also an excellent reason to suspect that the market may be able to internalize some small-scale, localized externalities.
- It won't help with large-scale, global externalities.



Public goods: in this course

- 7.1 Optimal Provision of Public Goods (quick recap)
- 7.2 Private Provision of Public Goods
- 7.3 Public Provision of Public Goods
- 7.4 Conclusion
- 9.1 Unanimous Consent on Public Goods Levels



Public Goods: A taxonomy

- Pure public goods: Goods that are perfectly non-rival in consumption and are non-excludable.
 - Non-rival in consumption: One individual's consumption of a good does not affect another's opportunity to consume the good.
 - Non-excludable: Individuals cannot deny each other the opportunity to consume a good.
- Impure public goods: Goods that satisfy the two public good conditions (non-rival in consumption and non-excludable) to some extent but not fully.

		Is the Good Rival in Consumption?: (Yes)	Is the Good Rival in Consumption?: (No)
Is the Good Excludable?	(Yes)	Private good (ice cream)	Impure public good (TV streaming)
Is the Good Excludable?	(No)	Impure public good (crowded city sidewalk)	Pure public good (national defense)



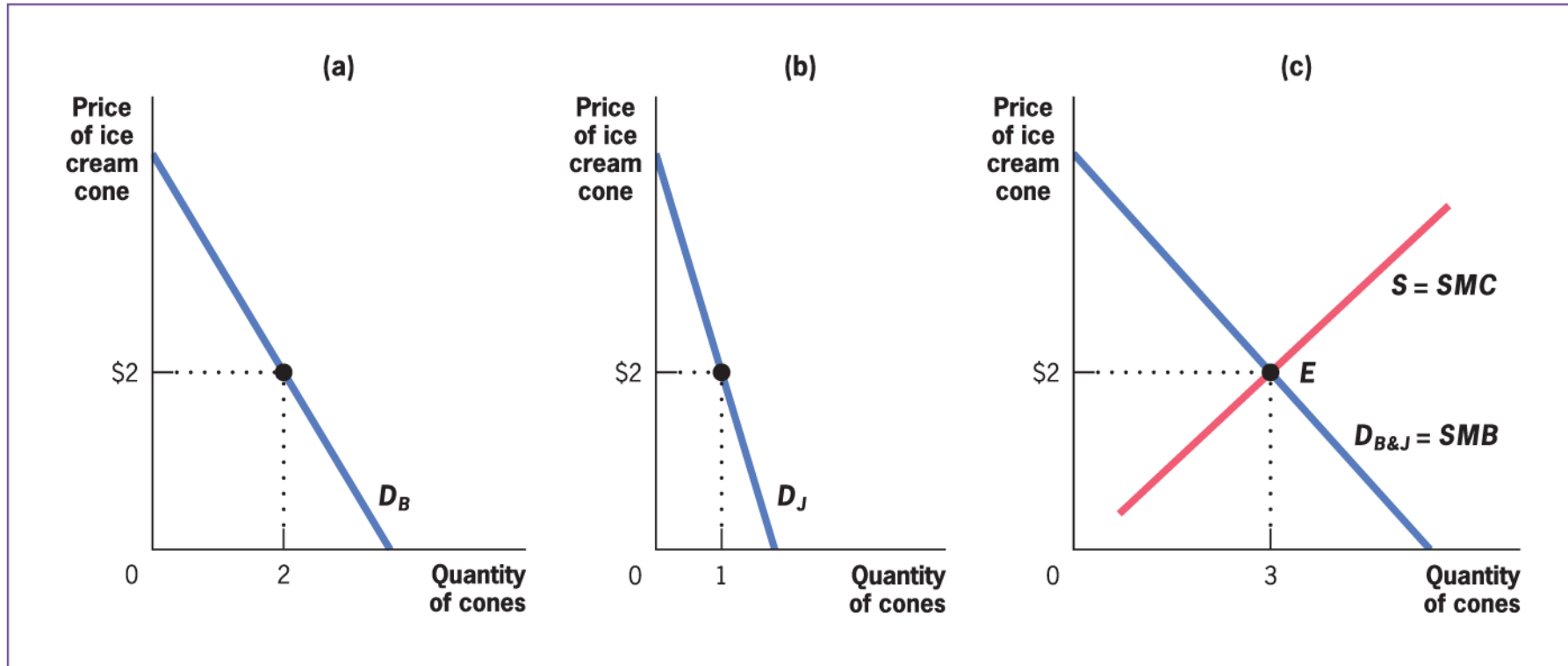
Optimal Provision of Public Goods: Introduction

- How much of the public good should society provide?
- Markets will not provide the correct amount!



Horizontal Summation in **Private** Goods Market

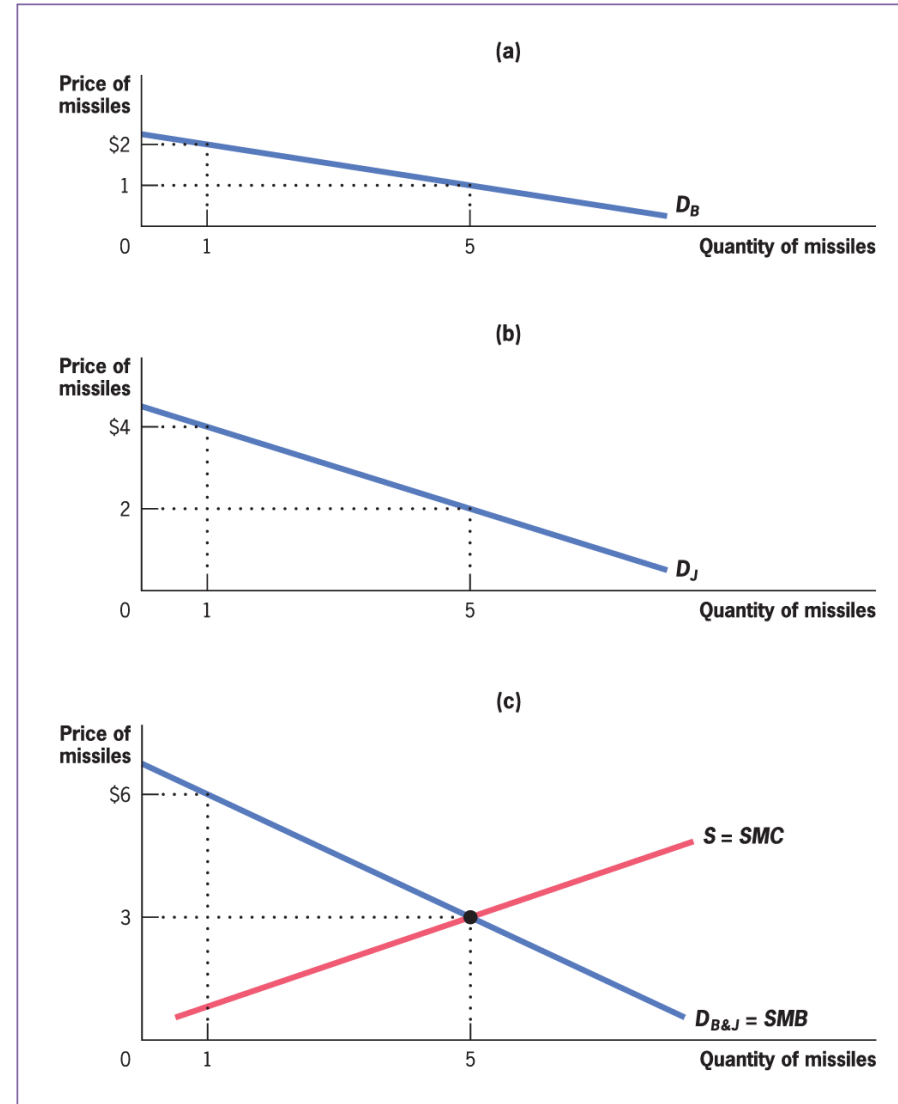
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Vertical Summation in **Public** Goods Market



Private Provision of Public Goods: Private-Sector Underprovision

- The market does not produce the efficient amount of public goods **because of the free rider problem.**
- Free rider problem: When an investment has a personal cost but a common benefit, individuals will underinvest.
- Since Ben's consumption of missiles also benefits Jerry, Jerry may not want to pay (or vice versa).
- This results in the private market producing an **inefficiently low quantity of the good.**



Can Private Providers Overcome the Free Rider Problem?

- The free rider problem does not lead to a complete absence of private provision of public goods.
- When private suppliers are given the ability to **overcome the problem of non-excludability**, they can produce the efficient quantity of the good.
- The private sector can in some cases combat the free rider problem to provide public goods by charging user **fees that are proportional to their valuation of the public good**.



Altruism and Social Capital

- Private markets provide public goods when people are altruistic.
- Altruistic: When individuals value the benefits and costs to others in making their consumption choices.
 - Many laboratory experiments provide evidence for altruism and show that people contribute to public goods.
- Social capital: The value of altruistic and communal behavior in society.
 - The quantity of social capital depends on how much people of the same community affected by the public good can “trust” each other and are therefore willing to chance their personal investment of time and effort into paying for the public good without any formal guarantee of reciprocity from other community members.



Warm Glow

- People might simply feel good about contributing to public goods or charity.
- Warm glow model: A model of the public goods provision in which individuals care about both the total amount of the public good and their particular contributions as well.
- Different from altruism because people don't care about just the amount of the public good.



Public Provision of Public Goods

- Even with private provision, there is a role for government provision of public goods.
- Under private provision, not everyone contributes to the good, even though everyone benefits.
- Government provision potentially solves the problem of noncontributors.
- Nonetheless, there are several challenges to government provision.



Private Responses to Public Provision: The Problem of Crowd-Out

- Crowd-out: As the government provides more of a public good, the private sector will provide less.
- In full crowd-out, government intervention accomplishes nothing because an increase of 5 units provided by the government results in a decrease of 5 units provided privately.
- Full crowd-out is rare. Partial crowd-out is much more common and can occur in two different cases:
 - When noncontributors to the public good are taxed to finance provision of the good
 - When individuals derive utility from their own contribution as well as from the total amount of public good



Contributors Versus Noncontributors

- Government provision is financed by payments by private individuals.
- By forcing noncontributors to contribute to the fund for public provision, the government can increase total public goods provision.
 - Noncontributors will be forced to increase their expenditures on the goods.
 - Contributors will experience an increase in effective wealth, which has a positive income effect on their purchase of the private good, so government intervention will not fully crowd out their spending.



Warm Glow and the Evidence on Crowd-Out

- If you get utility from your particular contributions for any reason (warm glow), then an increase in government contributions will not fully crowd out your giving.
 - If you only care about how much you give, government contributions have no effect on your giving.
- Existing evidence on crowd-out is quite mixed.
 - Studies assessing how individual contributions respond to government spending suggest a very small crowd-out.
 - These studies suffer from many bias problems.
 - Laboratory experiments suggest that crowd-out is large.
 - There is no consensus on the size of crowd-out.



EVIDENCE: Measuring Crowd-Out

- The evidence on crowd-out is mixed.
- Kingma (1989) looked at how contributions varied as local governments contributed different amounts to public radio.
- The study found that for every \$1 increase in government funding for public radio, private contributions fell by 13.5¢.
- Bias: Areas with high government contribution could be high income or have a high taste for radio.



The Right Mix of Public and Private

- One extreme is provision entirely by the public sector.
- The other extreme is subsidized or mandated private provision, with the government providing incentives.
- Contracting out: An approach through which the government retains responsibility for providing a good or service but hires private-sector firms to actually provide the good or service.
- Two problems with contracting out:
 - The private sector's incentives may not align with public goals, leading to lower public costs but worse outcomes along other dimensions that policy makers may care about.
 - Bidding in contracting out is often far from competitive.



Measuring the Costs and Benefits of Public Goods 1

- Optimal public good provision requires being able to measure both the benefits and the costs of providing public goods. In practice, this is quite difficult.
- Consider the case of a highway. Cost includes wages and materials.
 - What if, without this highway project, half of the workers on the project would be unemployed?
 - How can the government take into account that it is not only paying wages but also providing a new job opportunity for these workers?



Measuring the Costs and Benefits of Public Goods 2

- The benefits of highway construction are also difficult to measure.
 - What is the value of the time saved for commuters due to reduced traffic jams?
 - And what is the value to society of the reduced number of deaths if the highway is improved?
- These difficult questions are addressed by the field of cost-benefit analysis, which provides a framework for measuring the costs and benefits of public projects.



Unanimous Consent on Public Goods Levels (Ch 9.1)

Lindahl pricing: An approach to financing public goods in which individuals honestly reveal their willingness to pay and the government charges them that amount to finance the public good.

- Getting individuals to reveal their true marginal willingness to pay
- Aggregating these values to ensure that the social benefits exceed total cost
- Charging the individuals according to their willingness to pay



Lindahl Pricing 1

Lindahl's procedure operates as follows:

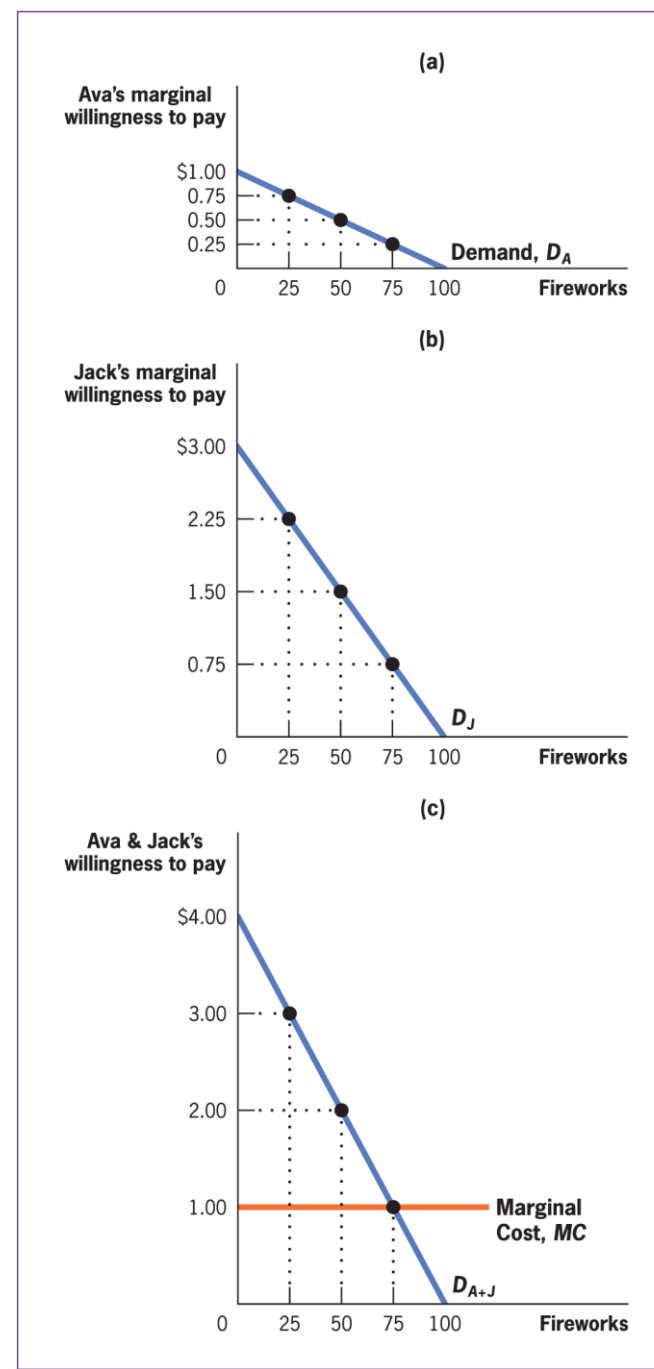
1. Announce tax prices for the public good.
2. Everyone says how much of the public good they want at those tax prices.
3. Repeat to construct a marginal willingness to pay schedule for each individual.

Recall that efficient provision requires that **total** marginal willingness to pay = MC .

4. Add up individual willingnesses to pay at each quantity of public good provided.
5. Find Q such that total marginal willingness to pay = MC .
6. Finance the public good by charging each individual their willingness to pay for that quantity.



Lindahl Pricing: Graphical Representation



Lindahl Pricing: Why It Works

Under Lindahl pricing, the government produces the efficient amount of the public good.

- This is because MC = total marginal willingness to pay.
- Each person's price is equal to their own marginal willingness to pay, so this is an equilibrium.
- Lindahl pricing also exemplifies *benefit taxation*.
- Benefit taxation: Taxation in which individuals are taxed for a public good according to their valuation of the benefit they receive from that good.



Problems with Lindahl Pricing

Lindahl pricing faces several problems that keep it from being used in practice:

1. Preference revelation problem: Individuals have an incentive to lie about their willingness to pay, to lower their price.
2. Preference knowledge problem: Individuals may not know their willingness to pay.
3. Preference aggregation problem: It is not obvious how to aggregate individual preferences into a social welfare function.



Exercises



MC 1

- According to the Coase theorem, the key issue in internalizing externalities is:
 - a) adding government intervention
 - b) addressing the free rider problem
 - c) defining property rights
 - d) removing transaction costs



MC 2

- Which of these does NOT reduce the problem of underprovision of public goods?
 - a) business improvement districts
 - b) government intervention
 - c) social capital
 - d) vertical summation



MC 3

• Which of these are issues in finding the optimal level of public provision of public goods?

I. measuring preferences

II. measuring costs

III. noncontributors

a) I & II only

b) I & III only

c) II & III only

d) I, II, & III



MC 4

Suppose that the government is providing a public good to two people, Lesedi and Sunil. To Lesedi, the marginal benefit of the public good is $MB = 1374 - 1,42Q$. Sunil's marginal benefit is $MB = 2287 - 1,53Q$. The marginal cost of providing the public good is constant and is equal to \$1596. What is the efficient quantity of the public good?

- a) 400
- b) 700
- c) 1000
- d) 1300



MC 5

In a private equilibrium, a positive production externality will lead to ____; a negative consumption externality will lead to ____.

- a) SMC less than PMC; SMC greater than PMC
- b) SMC less than PMC; SMC less than PMC
- c) SMB less than PMB; SMC greater than PMC
- d) SMC less than PMC; SMB less than PMB



True/False (please provide a justification)

- 1) If transaction costs are strictly positive, the Coase Theorem guarantees that private bargaining will still lead to the socially efficient outcome.
- 2) In the presence of a negative production externality, assigning property rights to the affected party (rather than the polluter) always leads to lower equilibrium production.
- 3) The free rider problem implies that private markets will never provide any public goods.



Exercises

EXERCISE 1

The care for public gardens entails a positive externality on the community. Assume that the marginal private cost can be represented by the following: $MC = 20 + 3Q$, where Q is the number of hours the gardener spends working. Assume also that the private benefit can be represented as follows: $Q = 70 - P$, and that the marginal external benefit from the gardening activity is 10.

- a) Please compute the equilibrium quantity achieved by the market
- b) Please compute the socially efficient quantity and the deadweight loss associated with part a



Exercises

EXERCISE 2

There are two types of citizens, type A and type B. Type A's demand for a public good is as follows: $P_A = 200 - Q$, while type B's demand is $P_B = 100 - 2Q$. Please draw these demand functions on two separate graphs.

Knowing that the marginal cost of providing the public good is 2250 and that there are 10 citizens per type (20 citizens total), please compute the optimal social quantity. Please provide a graphical representation of the equilibrium.

How much is the total cost for the good's provision? Can you think about a pricing system that allows the government to provide the optimal social quantity of the good? How much would type A citizens pay? And type B? What are the realization problems of this pricing system? Please discuss.

